**Question 7: [Programming, 10 Points]**

Consider the Poisson equation

$$\begin{aligned} -\Delta u &= f \quad \text{in } \Omega, \\ u &= g \quad \text{on } \Gamma, \end{aligned}$$

where $\Gamma = \partial\Omega$ denotes the boundary of the domain.
We have derived the nine-point stencil for solving the above PDE in the case $\Omega = [0,1]^2$ with uniform grid spacing.
The right-hand side $f$ and the boundary condition $g$ are chosen such that the exact solution is

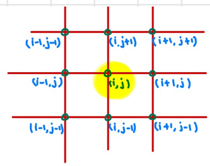
$$u(x,y) = x^4y^5 - 17\sin(xy).$$

From theory, we expect a convergence rate of order 4, which can be verified numerically. For bonus points, extend the implementation of the five-point stencil to the nine-point stencil and verify the order of convergence.

→ As, we know for 9-pt stencil from lecture notes,

should be $\oplus$ in lecture notes

$$\boxed{\begin{gathered} \mathcal{L}_h^9 \mathbf{u}_{i,j} = \mathbf{f}^{\times}_{i,j}, \\ \mathcal{L}_h^9 \mathbf{u}_{i,j} = \begin{cases} \mathbf{u}_{i,j}, & \text{if } i,j = 0 \text{ or } M, \\ \mathfrak{L}_h^9 \mathbf{u}_{i,j}, & \text{if } 1 \le i,j \le M-1, \end{cases} \quad \text{and} \quad \mathbf{f}^{\times}_{i,j} = \mathbf{f}_{i,j} + \frac{h^2}{12}\Delta \mathbf{f}_{i,j}, \end{gathered}}$$




$$\boxed{\mathfrak{L}_h^9 \mathbf{u}_{i,j} = \frac{1}{3h^2}\Big[10\mathbf{u}_{i,j} - \frac{1}{2}\left(\mathbf{u}_{i+1,j+1} + \mathbf{u}_{i+1,j-1} + \mathbf{u}_{i-1,j+1} + \mathbf{u}_{i-1,j-1}\right) - 2\left(\mathbf{u}_{i,j+1} + \mathbf{u}_{i,j-1} + \mathbf{u}_{i+1,j} + \mathbf{u}_{i-1,j}\right)\Big].}$$

So, for $j = 1$ & $i = 1, 2, .., M-1$, poisson eq$^n$ using above 9-pt stencil becomes,

$$\Rightarrow 20\,u_{i,1} - \left(u_{i+1,2} + u_{i+1,0} + u_{i-1,2} + u_{i-1,0}\right) - 4\left(u_{i,2} + u_{i,0} + u_{i+1,1} + u_{i-1,1}\right) = 6h^2 f^{*}_{i,1}$$

$$\Rightarrow 20\,u_{i,1} - u_{i+1,2} - u_{i-1,2} - 4u_{i,2} - 4u_{i+1,1} - 4_{i-1,1} = 6h^2 f^{*}_{i,1} + u_{i+1,0} + u_{i-1,0} + 4u_{i,0}$$

$$\Rightarrow \boxed{20u_{i,1} - u_{i+1,2} - u_{i-1,2} - 4u_{i,2} - 4u_{i+1,1} - 4_{i-1,1} = 6h^2 f^*_{i,1} + g_{i+1,0} + g_{i-1,0} + 4g_{i,0}} \quad \text{(i)}$$

For $j = 2$,

$$\Rightarrow \boxed{20u_{i,2} - (u_{i+1,3} + u_{i+1,1} + u_{i-1,3} + u_{i-1,1}) - 4(u_{i,3} + u_{i,1} + u_{i+1,2} + u_{i-1,2}) = 6h^2 f^*_{i,2}} \quad \text{(ii)}$$

For $j = M-1$,

$$\Rightarrow 20u_{i,M-1} - (u_{i+1,M} + u_{i+1,M-2} + u_{i-1,M} + u_{i-1,M-2}) - 4(u_{i,M} + u_{i,M-2} + u_{i+1,M-1} + u_{i-1,M-1}) = 6h^2 f^*_{i,M-1}$$

$$\Rightarrow 20u_{i,M-1} - u_{i+1,M-2} - u_{i-1,M-2} - 4u_{i,M-2} - 4u_{i+1,M-1} - 4u_{i-1,M-1} = 6h^2 f^*_{i,M-1} + u_{i+1,M} + u_{i-1,M} + 4u_{i,M}$$

$$\Rightarrow \boxed{20u_{i,M-1} - u_{i+1,M-2} - u_{i-1,M-2} - 4u_{i,M-2} - 4u_{i+1,M-1} - 4u_{i-1,M-1} = 6h^2 f^*_{i,M-1} + g_{i+1,M} + g_{i-1,M} + 4g_{i,M}} \quad \text{(iii)}$$

$\Rightarrow$ (i) in matrix form,

$$\boxed{\begin{bmatrix} -4 & -1 \end{bmatrix} \begin{bmatrix} u_{i-1,1} \\ u_{i-1,2} \end{bmatrix} + \begin{bmatrix} 20 & -4 \end{bmatrix} \begin{bmatrix} u_{i,1} \\ u_{i,2} \end{bmatrix} + \begin{bmatrix} -4 & -1 \end{bmatrix} \begin{bmatrix} u_{i+1,1} \\ u_{i+1,2} \end{bmatrix} = 6h^2 f^*_{i,1} + g_{i+1,0} + g_{i-1,0} + 4g_{i,0}} \quad \text{(iv)}$$

$\Rightarrow$ (ii) in matrix form,

$$\boxed{\begin{bmatrix} -1 & -4 & -1 \end{bmatrix} \begin{bmatrix} u_{i-1,1} \\ u_{i-1,2} \\ u_{i-1,3} \end{bmatrix} + \begin{bmatrix} -4 & 20 & -4 \end{bmatrix} \begin{bmatrix} u_{i,1} \\ u_{i,2} \\ u_{i,3} \end{bmatrix} + \begin{bmatrix} -1 & -4 & -1 \end{bmatrix} \begin{bmatrix} u_{i+1,1} \\ u_{i+1,2} \\ u_{i+1,3} \end{bmatrix} = 6h^2 f^*_{i,2}} \quad \text{(v)}$$

$\Rightarrow$ (iii) in matrix form

$$\boxed{[-1\ \ -4]\begin{bmatrix} u_{i-1,M-2} \\ u_{i-1,M-1} \end{bmatrix} + [-4\ \ 20]\begin{bmatrix} u_{i,M-2} \\ u_{i,M-1} \end{bmatrix} + [-1\ \ -4]\begin{bmatrix} u_{i+1,M-2} \\ u_{i+1,M-1} \end{bmatrix} = 6h^2 f^*_{i,M-1} + g_{i+1,M} + g_{i-1,M} + 4g_{i,M}} \quad \text{— (vi)}$$

From (iv), (v) & (vi), we can write,

$$\boxed{-D_1\,\bar{U}_{i-1} + D_2\,\bar{U}_i - D_1\,\bar{U}_{i+1} = \bar{f}_i} \quad \text{— (vii)} \quad \leftarrow \text{eq}^n \text{ for fix — } i^{\text{th}} \text{ grid}$$

where,

$$\bar{U}_i = \begin{bmatrix} u_{i,1} \\ u_{i,2} \\ \vdots \\ u_{i,M-1} \end{bmatrix}_{(M-1)\times 1}, \quad \text{similarly, } \bar{U}_{i-1} \ \& \ \bar{U}_{i+1}$$

& $$D_1 = \begin{bmatrix} 4 & 1 & 0 & \cdots & \cdots & 0 \\ 1 & 4 & 1 & 0 & \cdots & 0 \\ 0 & \ddots & & & & \vdots \\ \vdots & & \ddots & & & 1 \\ 0 & \cdots & \cdots & 0 & 1 & 4 \end{bmatrix}_{(M-1)\times(M-1)} \qquad D_2 = \begin{bmatrix} 20 & -4 & 0 & \cdots & 0 \\ -4 & 20 & -4 & 0 \cdots & 0 \\ 0 & \ddots & & & \vdots \\ \vdots & & \ddots & & -4 \\ 0 & \cdots & 0 & -4 & 20 \end{bmatrix}_{(M-1)\times(M-1)}$$

& $$\bar{f}_i = \begin{bmatrix} 6h^2 f^*_{i,1} + g_{i+1,0} + g_{i-1,0} + 4g_{i,0} \\ 6h^2 f^*_{i,2} \\ \vdots \\ 6h^2 f^*_{i,M-2} \\ 6h^2 f^*_{i,M-1} + g_{i+1,M} + g_{i-1,M} + 4g_{i,M} \end{bmatrix}_{(M-1)\times 1}$$

$\Rightarrow$ for $i = 1$, (vii) becomes,

$$-D_1 \bar{u}_0 + D_2 \bar{u}_1 - D_1 \bar{u}_2 = \bar{f}_1$$

Since, $\bar{u}_0 = [u_{0,1} \quad u_{0,2} \quad \cdots \quad u_{0,M-1}]^T$
$= [g_{0,1} \quad g_{0,2} \quad \cdots \quad g_{0,M-1}]^T = \bar{g}_0$

$$\therefore \quad \boxed{D_2 \bar{u}_1 - D_1 \bar{u}_2 = \bar{f}_1 + D_1 \bar{g}_0}$$

$\Rightarrow$ for $i=2$,

$$\boxed{-D_1 \bar{u}_1 + D_2 \bar{u}_2 - D_1 \bar{u}_3 = \bar{f}_2}$$

$\Rightarrow$ for $i = M-1$,

$$-D_1 \bar{u}_{M-2} + D_2 \bar{u}_{M-1} - D_1 \bar{u}_M = \bar{f}_{M-1}$$

Since, $\bar{u}_M = [u_{M,1} \quad u_{M,2} \quad \cdots \quad u_{M,M-1}]^T$
$= [g_{M,1} \quad g_{M,2} \quad \cdots \quad g_{M,M-1}]^T = \bar{g}_M$

$$\therefore \quad \boxed{-D_1 \bar{u}_{M-2} + D_2 \bar{u}_{M-1} = \bar{f}_{M-1} + D_1 \bar{g}_M}$$

Now stacking eq$^{NS}$ $\forall\ i = 1, 2, \ldots, M-1$

$$\boxed{A \bar{u} = \bar{f}}$$

where,

$$A = \begin{bmatrix} D_2 & -D_1 & 0 & \cdots & 0 \\ -D_1 & D_2 & -D_1 & 0 \cdots & 0 \\ 0 & & \ddots & & \vdots \\ \vdots & & & \ddots & -D_1 \\ 0 & \cdots & 0 & -D_1 & D_2 \end{bmatrix}_{(M-1)^2 \times (M-1)^2}, \quad \bar{u} = \begin{bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ \vdots \\ \bar{u}_{M-1} \end{bmatrix}_{(M-1)^2 \times 1}, \quad \bar{f} = \begin{bmatrix} \bar{f}_1 + D_1 \bar{g}_0 \\ \bar{f}_2 \\ \vdots \\ \bar{f}_{M-2} \\ \bar{f}_{M-1} + D_1 \bar{g}_M \end{bmatrix}_{(M-1)^2 \times 1}$$

Let $\quad A = A_1 + A_2$

s.t., $\quad A_1 = \underbrace{I \otimes D_2}_{(M-1)\times(M-1)\text{ matrices}} = \begin{bmatrix} 1 & 0 & \cdots & & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ & & & \ddots & 0 \\ 0 & \cdots & & 0 & 1 \end{bmatrix} \otimes D_2$

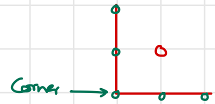
$$A_1 = \begin{bmatrix} D_2 & 0 & \cdots & & 0 \\ 0 & D_2 & 0 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & \cdots & 0 & D_2 \end{bmatrix}_{(M-1)^2 \times (M-1)^2}$$

& $\quad A_2 = \bar{I} \otimes D_1 = \begin{bmatrix} 0 & -1 & 0 & \cdots & \cdots & 0 \\ -1 & 0 & -1 & 0 & \cdots & 0 \\ 0 & & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & & 0 \\ & & & & & -1 \\ 0 & \cdots & \cdots & 0 & -1 & 0 \end{bmatrix} \otimes D_1$

$$\therefore A_2 = \begin{bmatrix} 0 & -D_1 & 0 & \cdots & & 0 \\ -D_1 & 0 & -D_1 & 0 & \cdots & 0 \\ \vdots & & \ddots & \ddots & & \vdots \\ & & & \ddots & & 0 \\ & & & & & -D_1 \\ 0 & \cdots & \cdots & 0 & -D_1 & 0 \end{bmatrix}_{(M-1)^2 \times (M-1)^2}$$

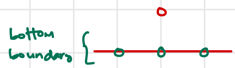
& $\quad \bar{f} = \begin{bmatrix} \bar{f}_1 + D_1 \bar{g}_0 \\ \bar{f}_2 \\ \vdots \\ \bar{f}_{M-2} \\ \bar{f}_{M-1} + D_1 \bar{g}_M \end{bmatrix}_{(M-1)^2 \times 1} \implies D_1 \bar{g}_0 = \begin{bmatrix} 4 & 1 & 0 & \cdots & & 0 \\ 1 & 4 & 1 & 0 & \cdots & 0 \\ 0 & & \ddots & \ddots & & \vdots \\ \vdots & & & & & 0 \\ & & & & 4 & 1 \\ 0 & \cdots & \cdots & & 1 & 4 \end{bmatrix} \begin{bmatrix} g_{0,1} \\ g_{0,2} \\ \vdots \\ \vdots \\ g_{0,M-1} \end{bmatrix} = \begin{bmatrix} 4 g_{0,1} + g_{0,2} \\ g_{0,1} + 4 g_{0,2} + g_{0,3} \\ g_{0,2} + 4 g_{0,3} + g_{0,4} \\ \vdots \\ g_{0,M-3} + 4 g_{0,M-2} + g_{0,M-1} \\ g_{0,M-2} + 4 g_{0,M-1} \end{bmatrix}_{(M-1) \times 1}$

Similarly $$D_1 \bar{g}_M = \begin{bmatrix} 4g_{M,1} + g_{M,2} \\ g_{M,1} + 4g_{M,2} + g_{M,3} \\ g_{M,2} + 4g_{M,3} + g_{M,4} \\ \vdots \\ g_{M,M-3} + 4g_{M,M-2} + g_{M,M-1} \\ g_{M,M-2} + 4g_{M,M-1} \end{bmatrix}_{(M-1)\times 1}$$




⇒ Inside pt. near corner has 5− boundary neighbour.

Other inside pts. near boundary has 3− boundary neighbour.




And $\bar{f}_i$ $(i = 1, 2, \ldots, M-1)$ are defined earlier.

Therefore,

$$\bar{f} = \begin{array}{r|l} \bar{f}_1 + D_1\bar{g}_0 & \begin{array}{l} 6h^2 f^*_{1,1} + (4g_{0,1} + g_{0,2}) + (g_{0,0} + 4g_{1,0} + g_{2,0}) \\ 6h^2 f^*_{1,2} + (g_{0,1} + 4g_{0,2} + g_{0,3}) \\ \vdots \\ 6h^2 f^*_{1,M-1} + (g_{0,M-2} + 4g_{0,M-1}) + (g_{0,M} + 4g_{1,M} + g_{2,M}) \end{array} \\ \hline \bar{f}_2 & \begin{array}{l} 6h^2 f^*_{2,1} + (g_{1,0} + 4g_{2,0} + g_{3,0}) \\ 6h^2 f^*_{2,2} \\ \vdots \\ 6h^2 f^*_{2,M-1} + (g_{1,M} + 4g_{2,M} + g_{3,M}) \end{array} \\ \hline & \vdots \\ \hline \bar{f}_{M-1} + D_1\bar{g}_M & \begin{array}{l} 6h^2 f^*_{M-1,1} + (4g_{M,1} + g_{M,2}) + (g_{M-2,0} + 4g_{M-1,0} + g_{M,0}) \\ 6h^2 f^*_{M-1,2} + (g_{M,1} + 4g_{M,2} + g_{M,3}) \\ \vdots \\ 6h^2 f^*_{M-1,M-1} + (g_{M,M-2} + 4g_{M,M-1}) + (g_{M-2,M} + 4g_{M-1,M} + g_{M,M}) \end{array} \end{array}$$

⇒ Also exact sol. is given, $u = x^4 y^5 - 17\sin(xy)$

$\therefore \; \Delta u = 20x^4y^3 + 12x^2y^5 + 17x^2\sin(xy) + 17y^2\sin(xy)$

As, $f = -\Delta u \Rightarrow \Delta f = -120x^4 y - 480x^2y^3 - 24y^5 - 136\,xy\cos(xy) - 68\sin(xy) + 17x^4\sin(xy) + 34x^2y^2\sin(xy) + 17y^4\sin(xy)$

$$\therefore \left\{ f^*_{i,j} = f_{i,j} + \frac{h^2}{12}\Delta f_{i,j} = -(\Delta u)_{i,j} - (\Delta^2 u)_{i,j}\frac{h^2}{12} \right\}$$

⇒ See Python Notebook